QUANT

Number System

(Digital Root, Unit's Digit, Last two digits
& other important concepts)

Assignment 04

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Numbers Online Assignment -04

1. A, B, C are three distinct digits. AB is a two digit number and CCB is a three digit number such that $(AB)^2$ = CCB where CCB > 320. What is the possible value of the digit B?

1. 1 2. 0 3. 3 4. 9

2. What is the unit's digit of the following expression? $125^{301} \times 74^{72} - 187^{21}$.

1.2 2. 3 3. 4 4. None of these

3. $2^{32} + 2^{33} + 2^{34} + 2^{35}$ is divisible by

1. 7 2. 9 3. 15 4. 25

4. What is the digit in the unit's place of 2^{51}?

1. 2 2. 8 3. 1 4. 4

5. Find the unit's digit of the expression $11^1 \times 12^2 \times 13^3 \times 14^4 \times 15^5 \times 16^6$?

1. 1 2. 9 3. 7 4. 0

6. $K = 1272^{239} \times 976^{511} - 279^{10241}$. What will be the last digit of K?

1. 7 2. 3 3. 9 4. 1

7. G(x) = Sum of all the digits of natural number x, e.g. G(182) = 1 + 8 + 2 = 11. What is the value of P? where P = {G(63) + G(64) + G(65) + …….. + G(187)}

1. 1184 2. 1279 3.1176 4. 1224

8. Find the unit's digit of the expression $11^1 + 12^2 + 13^3 + 14^4 + 15^5 + 16^6$.

1. 1 2. 9 3. 7 4. 0

9. What will be the two right most digits of the number 3^{132}?

1. 21 2. 31 3. 41 4. 61

10. What will be the last two digits of the number 2^{72}?

1. 76 2. 96 3. 56 4. 46

Answer keys & Explanations

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1.	1	The only number satisfying this condition is 21. As 21 × 21 = 441, so possible value of B is 1.
2.	2	Take only the unit's digit of all the numbers i.e. $5^{301} \times 4^{72} - 7^{21}$. Now divide every power by 4 and take the remainder power. The remaining powers are $5^1 \times 4^0 - 7^1 \Rightarrow 5 \times 6 - 7 = 3$ is the unit's digit (Note: do not think, the remainder power in case of 4 is 0, so 4^0 will be equal to 1. But remainder power 0 implies that the last digit of the 4th power.)
3.	3	Take 2^{32} common → 2^{32} (1 + 2 + 4 + 8) → 2^{32} (15). Thus the number is divisible by 15. Thus 3rd option.
4.	2	Divide 51 by cyclicity of 2 i.e. 4. Remainder = 3. Now you can find $2^3 = 8$. Thus 2nd option.
5.	4	Due to availability of an even number and 15^5, the unit digit of the given expression would be zero.
6.	3	As the unit's digit is the only consideration. 2^{239} the power 239 ÷ 4 = 3 is the remainder 2^3 = 8, 8 is the unit's digit of first term. Any power of 6 always results in a 6. Lastly 9^{10241} the power 10241 ÷ 4 = 1 is the remainder $9^1 = 9$. Hence the final unit's digit will be 8 × 6 – 9 = 39, hence 9 will be the unit's digit. That is given in the 3rd option and hence is the answer.
7.	2	At the hundred's place 1 will come 88 times. Thus the total at the hundred's place will be 1 × 88 = 88. At ten's place starting from 63 upto 162, every digit will appear 10 times as these are 100 consecutive numbers. Besides 6 will appear 7 more times (from 163 – 169), 7 will appear 10 times (170 – 179) and 8 will appear 8 times (180 – 187). The sum at the ten's place will be Σ9 × 10 + 6 ×7 + 7 × 10 + 8 × 8 = 626. At unit's place starting from 63 upto 162, every digit will appear 10 times as these are 100 consecutive numbers. Besides you have unit's digit of numbers from 163 – 187. Their sum will be (3 + 4+ 5 + 6 + 7+ 8 + 9) + Σ9 + Σ7 ⇒The total sum thus be Σ9 × 10 + 42 + Σ9 + Σ7 = 450 + 42 + 45 + 28 = 565. Now the sum of all the digits will be 88 + 626 + 565 = 1279. **OR** We can check the digital root of the series, (as discussed in the video of **DIGITAL ROOT**)
8.	2	Final unit digit of this expression would be 1 + 4 + 7 + 6 + 5 + 6 i.e 9.
9.	3	$3^{132.}$ We know that cyclicity of 3 is 20. ∴ We divide 132 by 20 and find the remainder which comes out to be 12. Hence 12th number in the sequence of powers of 3 is 41. So answer is 41 which is given as 3rd option.
10.	2	2^{72}. Using the cyclicity of 2 which is of 20, we need to find the answer. Whenever we have to find the last 2 digits of a number with base 2, we subtract 21 from the power and then we divide the power by 20 and find the remainder. So 72 – 21 = 51 divided by 20 gives 11 as remainder. So 12th number in the sequence of powers of 2 is 96. So answer is 96 which is given as 2nd option.